University of Sheffield

COM3505
The Internet of Things

Assignment Brief: ESP32 IoT Device with Sensor and LED Control via Web Interface

Pair Work Assignment

Coursework Introduction

Assignment Overview

Design and implement a complete IoT system using the ESP32 microcontroller.
This coursework is a collaborative project to be completed in pairs.

Core Tasks & Objectives

- Read data from at least one sensor (e.g., temperature, light)
- Control 3 or more LEDs with dynamic animation patterns
- Connect the ESP32 to a local Wi-Fi network
- Send sensor data to a Python Flask server
- Display live data and controls on a web browser interface

Hardware Kit

Each pair will be provided with a complete kit containing all necessary components to build the IoT device.

Component List

- ESP32 Feather Board + Stacking Headers
- Breadboard
- LEDs: 5× Red, 5× Green, 5× Yellow (3mm)
- "Workstation 1" LED Kit (PCB)
- Resistors: 120Ω (×12), 180Ω, 4.7kΩ, 220kΩ, 1MΩ
- Push Button Switch + 10-way Ribbon Cable
- USB-C Cable + Plastic Storage Box

Functional Requirements

1 Sensor Integration

Must implement at least one physical sensor (e.g., temperature, light, motion).

Data should be read and transmitted to the server every 2–5 seconds consistently.

2 LED Patterns

Control 3 or more LEDs with distinct animation animation patterns.

Required patterns: Solid Colour, Rainbow, Chase effect, and Fire/Flame simulation.

3 Wi-Fi Connectivity

ESP32 must successfully connect to a local Wi-Fi network.

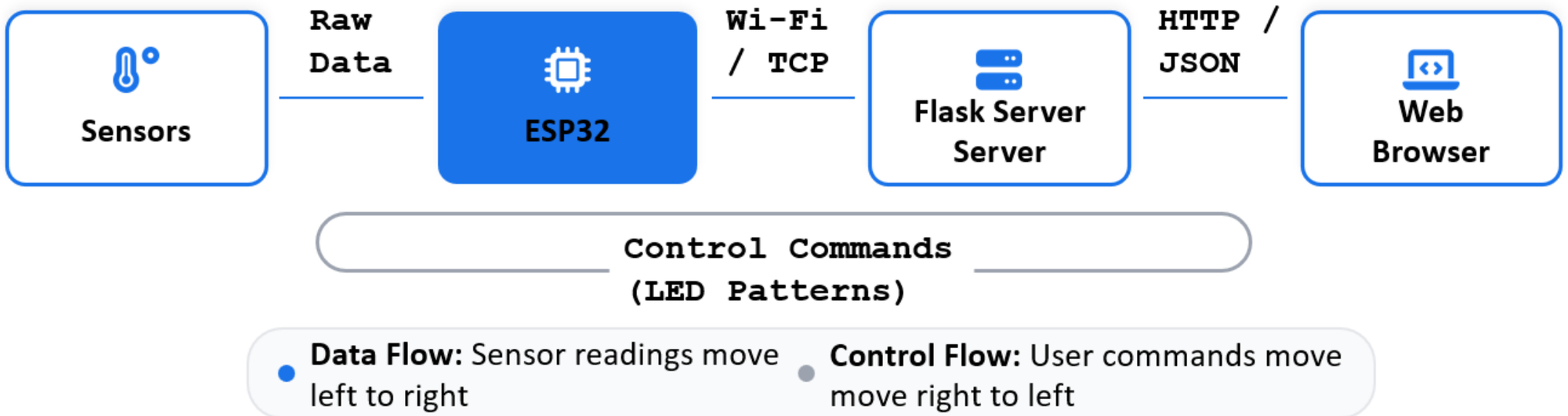
Device IP address must be printed to the Serial Monitor upon successful connection.

4 Web Interface (Flask)

Python Flask server hosting a live-updating web page (AJAX).

Interface must display real-time sensor data and provide buttons to control LED patterns remotely.

System Architecture and Software Stack



Layer	Technology
Microcontroller IDE	Arduino IDE / PlatformIO
ESP32 Language	C++ (Arduino Framework)
Web Server	Python + Flask
Frontend	HTML / CSS / JavaScript (AJAX)
Protocol	HTTP / REST

Deliverables

To complete the assessment, your pair must submit the following three items via the university portal by the assigned deadline.

1. **Code Submission**

   Complete ESP32 source code (C++/Arduino) and the Python Flask server code.

2. **Project Report**

   A maximum 4-page document including hardware diagrams, system architecture, and integration explanation.

3. **Demonstration Video**

   Visual evidence of Wi-Fi connection establishment, live sensor data updates, and LED patterns being controlled via the browser.

Assessment Criteria

Criteria	Marks
Hardware integration (sensor + LEDs)	15
LED patterns and animation logic	20
ESP32 Wi-Fi connectivity	15
Web interface design and functionality	20
Data communication (sensor → server → web)	15
Code quality, comments, and structure	10
Team collaboration & report	5
Total	100

Grand Bands

Mark Range	Description
40–49	• Fail / Minimal effort Project meets only very basic requirements. Sensor readings may not display, LEDs may not work, Wi-Fi may not function, web interface is missing or broken, patterns not implemented. Poor documentation.
50–59	• Pass / Satisfactory Basic functionality works: ESP32 connects to Wi-Fi, sensor readings display, at least 3 LEDs operate with simple control. Web interface may exist but is minimal, and LED patterns may be very basic (solid color only). Documentation covers setup but is incomplete.
60–69	• Credit / Good Project meets most requirements: sensor readings are live, LEDs can be controlled, at least one pattern implemented beyond solid color. Web interface shows sensor data and allows LED control. Code is reasonably structured with comments, documentation is clear.
70–79	• Distinction / Very Good Project fully meets requirements: multiple LED patterns implemented (rainbow, chase, fire), sensor readings update in real time, web interface is user-friendly, and communication between ESP32 and Python server is robust. Code is well-organized, documented, and readable. Documentation is complete and professional.
80–100	• High Distinction / Excellent Project exceeds requirements: creative LED patterns, optional features implemented (e.g., auto pattern changes based on sensor thresholds, graphs for sensor data), mobile-friendly web interface, robust error handling, highly readable and structured code. Excellent teamwork and professional-quality documentation.